

Further support

AQA supports the centre assessed units in a number of ways.

- AQA holds annual standardising meetings on a regional basis for all centre assessed components. Section 6 of this specification provides further details about these meetings
- A Teacher Resource Bank which includes further information and guidance from the Principal Moderator.
- Assessment Advisers are appointed by AQA to provide advice on centre assessed units. Every centre is allocated an Assessment Adviser. Details are sent to the Head of Department. The Assessment Advisers can provide guidance on
 - issues relating to the carrying out of assignments for assessment
 - application of marking guidelines
 - administrative issues related to the centre assessed units.

3.8.4 General Marking Guidance for each PSA

Centres should bear in mind that satisfactory completion of a PSA task by the candidate should be judged in the context of an ability to work safely and in an organised manner, when demonstrating appropriate manipulative skills.

Each task should be graded on a three point scale (0, 1 or 2 marks) with the following general guidelines for the award of each point on the scale.

Assessment Descriptors

2 marks	Candidates are able to follow a set of instructions for the task in a safe and organised way. Measurements are precise and within the expected range. Candidates require minimal additional guidance to carry out the task in a competent manner and are able to produce an outcome which is within the expected tolerance for the activity or produce a set of results, most of which are correct.
1 mark	Candidates are able to follow a set of instructions for the task in a reasonably safe way, but could be better organised. Measurements are imprecise or outside the expected range. Candidates require some additional guidance to carry out the task to a standard which is considered appropriate and produce an outcome, which, whilst acceptable, may not be within the expected tolerance for the activity or produce a set of results, only some of which are correct.
0 marks	Candidates have significant difficulty in following a set of instructions for the task and their work is poorly organised or unsafe. Measurements are imprecise or outside the expected range. Candidates require significant additional guidance to carry out the task to a standard which is considered appropriate and produce an outcome which is significantly outside the expected tolerance for the activity or produce a set of results, few of which are correct. The following sections detail, for AS and A2, possible contexts and marking guidance for each area of chemistry.

Practical Skills Assessment: AS Inorganic Chemistry

Task and possible context	Specific marking guidance
Make up a volumetric solution For example: The preparation of a standard solution of sodium carbonate	2 marks: All areas of the task are carried out competently. The weighing is precise and within the required range. The transfer of solid to a graduated flask is done with care. The solution is made up to the mark, accurately. 1 mark: One of the areas of the task is performed poorly. The weighing is imprecise or outside the required range OR The transfer of solid to the graduated flask is careless OR The solution is made up inaccurately. (e.g. the flask is over-filled) 0 marks: At least two of the areas of the task are performed poorly. The weighing is imprecise or outside the required range. The transfer of solid to the graduated flask is careless. The solution is made up inaccurately.
Carry out a simple acid-base titration For example: Determine the concentration of unknown hydrochloric acid by titration	2 marks: All areas of the task are carried out competently. The burette is filled safely with the correct reagent (including below the tap). The pipette and filler, burette and conical flask are all used correctly. The titration results are concordant and the average titre is judged accurate. 1 mark: One of the areas of the task is performed poorly. The burette is filled with the incorrect reagent or the funnel is left in or the burette is not filled below the tap OR One of either, the pipette, pipette filler, burette or conical flask is not used correctly OR The titration results are not concordant or the average titre is inaccurate. 0 marks: At least two of the areas of the task are performed poorly. The burette is filled with the incorrect reagent or the funnel is left in or the burette is not filled below the tap. One of either, the pipette, pipette filler, burette or conical flask is not used correctly. The titration results are not concordant or the average titre is inaccurate.
Carry out some inorganic tests For example: Tests for anions	2 marks: All areas of the task are carried out competently. The quantities of reagents are appropriate. The tests (heating, shaking etc.) are carried out safely and with due care. Most of the observations are correct. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate OR The tests (heating, shaking etc.) are carried out in a careless manner OR Only some of the observations are correct. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate. The tests (addition, heating, shaking etc.) are carried out in a careless manner. Few of the observations are correct.

Practical Skills Assessment: AS Physical Chemistry

Task and possible context	Specific marking guidance
Measure an enthalpy change For example: Use Hess's law to find an unknown enthalpy change, such as the reaction of anhydrous copper(II) sulfate with water to produce hydrated crystals	2 marks: All areas of the task are carried out competently. Masses and volumes are measured precisely and within the required range. Initial/final temperatures are measured precisely and mixing is complete. The results lead to an enthalpy change which is within the expected range. 1 mark: One of the areas of the task is performed poorly. Masses or volumes are measured imprecisely or not in the required range OR Temperatures are measured imprecisely or mixing is incomplete OR The results lead to an enthalpy change which is outside the expected range. 0 marks: At least two of the areas of the task are performed poorly. Masses or volumes are measured imprecisely or not in the required range. Temperatures are measured imprecisely or mixing is incomplete. The results lead to an enthalpy change which is outside the expected range.
Determine the M_r of a volatile liquid or the M_r of a gas For example: Determine the M_r of hexane or the M_r of carbon dioxide	2 marks: All areas of the task are carried out competently. The apparatus is weighed precisely and handled carefully. The transfer of liquid or gas is carried out safely and with due care. The apparatus is equilibrated and all necessary measurements taken. 1 mark: One of the areas of the task is performed poorly. The apparatus is weighed imprecisely or handled without due care OR The transfer of liquid or gas is not carried out safely or with due care OR The apparatus has not equilibrated or some measurements are not taken. 0 marks: At least two of the areas of the task are performed poorly. The apparatus is weighed imprecisely or handled without due care. The transfer of liquid or gas is not carried out safely or with due care. The apparatus has not equilibrated or some measurements are not taken.
Investigate how the rate of a reaction changes with temperature For example: Investigate the rate of reaction of sodium thiosulfate with acid at different temperatures	2 marks: All areas of the task are carried out competently. The quantities of reagents are appropriate and the apparatus is safe. Heating is carried out with due care and only as long as necessary. The change in the measured rate is within the expected range. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate or the apparatus is unsafe OR Heating is carried out with insufficient care or longer than necessary OR The change in the measured rate is not within the expected range. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate or the apparatus is unsafe. Heating is carried out with insufficient care or longer than necessary. The change in the measured rate is not within the expected range.

Practical Skills Assessment: AS Organic Chemistry

Task and possible context	Specific marking guidance
Distil a product from a reaction For example: The preparation of ethanal from the oxidation of ethanol or the preparation of cyclohexene from cyclohexanol	2 marks: All areas of the task are carried out competently. The apparatus set-up is safe and appropriate. Heating is carried out with due care and only as long as necessary. The yield of product is appropriate. 1 mark: One of the areas of the task is performed poorly. The apparatus set-up is inappropriate OR Heating is carried out with insufficient care or longer than necessary OR The yield of product is inappropriate. 0 marks: At least two of the areas of the task are performed poorly. The apparatus set-up is inappropriate. Heating is carried out with insufficient care or longer than necessary. The yield of product is inappropriate.
Carry out some organic tests For example: Tests for alkene, alcohol, acid, aldehyde	2 marks: All areas of the task are carried out competently. The quantities of reagents are appropriate. The tests (heating, shaking etc.) are carried out safely and with due care. Nearly all of the observations are correct. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate OR The tests (heating, shaking etc.) are carried out in a careless manner OR Only some of the observations are correct. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate. The tests (addition, heating, shaking etc.) are carried out in a careless manner. Few of the observations are correct.
Investigate the combustion of alcohols For example: Use a calorimetric method to measure the enthalpies of combustion in an homologous series	2 marks: All areas of the task are carried out competently. Masses and volumes are measured precisely and within the required range. Initial/final temperatures are measured precisely. The range and trend in enthalpies is as expected for the series. 1 mark: One of the areas of the task is performed poorly. Masses or volumes are measured imprecisely or not in the required range OR Temperatures are measured imprecisely OR The range or trend in enthalpies is not as expected for the series. 0 marks: At least two of the areas of the task are performed poorly. Masses or volumes are measured imprecisely or not in the required range. Temperatures are measured imprecisely. The range or trend in enthalpies is not as expected for the series.

Practical Skills Assessment: A2 Inorganic Chemistry

Task and possible context	Specific marking guidance
Carry out a redox titration For example: The analysis of iron tablets by titration using acidified potassium manganate(VII)	2 marks: All areas of the task are carried out competently. The burette is filled safely with the correct reagent (including below the tap) The pipette and filler, burette and conical flask are all used correctly. The titration results are concordant and the average titre is accurate. 1 mark: One of the areas of the task is performed poorly. The burette is filled with the incorrect reagent or the funnel is left in or the burette is not filled below the tap OR One of either the pipette, pipette filler, burette or conical flask is used incorrectly OR The titration results are not concordant or the average titre is inaccurate. 0 marks: At least two of the areas of the task are performed poorly. The burette is filled with the incorrect reagent or the funnel is left in or the burette is not filled below the tap. One of either the pipette, filler, burette or conical flask is used incorrectly. The titration results are not concordant or the average titre is inaccurate.
Investigate the chemistry of transition metal compounds in a series of experiments For example: The chemistry of copper compounds	2 marks: All experiments are carried out competently. The quantities of reagents are appropriate. All experiments are carried out safely and with due care. Nearly all of the observations are correct. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate OR Some of the experiments are carried out in a careless manner OR Only some of the observations are correct. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate. Many of the experiments are carried out in a careless manner. Few of the observations are correct.
Prepare an inorganic complex For example: The preparation of hexaamminecobalt(III) chloride	2 marks: All areas of the task are carried out competently. The quantities of reagents are appropriate for the preparation. The apparatus set-up for the preparation is safe and appropriate. The experiment is carried out safely and produces an appropriate quantity and quality of product. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate for the preparation OR The apparatus set-up for each experiment is unsafe or inappropriate OR The experiments are carried out with insufficient care or the yield is poor. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate for the preparation. The apparatus set-up for each experiment is unsafe or inappropriate. The experiments are carried out with insufficient care or the yield is poor.

Practical Skills Assessment: A2 Physical Chemistry

Task and possible context	Specific marking guidance
Carry out a kinetic study to determine the order of a reaction For example: An iodine clock experiment e.g. the reaction of sulfite ions with iodate(V) ions	2 marks: All areas of the task are carried out competently. The quantities of reagents are measured precisely. Times are measured accurately and recorded precisely. Sufficient values are on a good straight line and the order of reaction is in the expected range. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are measured imprecisely OR Times are measured inaccurately or recorded imprecisely OR The values are scattered or the order is not in the expected range. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are measured imprecisely. Times are measured inaccurately or recorded imprecisely. The values are scattered or the order is not in the expected range.
Determine an equilibrium constant For example: Determine a value of K_c for the reaction of ethanol with ethanoic acid	2 marks: All areas of the task are carried out competently. The quantities of reagents are measured precisely. The titrations are carried out with due care and data recorded precisely. The value of the equilibrium constant is in the expected range. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are measured imprecisely OR Titrations are carried out with insufficient care or data recorded imprecisely OR The value of the equilibrium constant is not in the expected range. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are measured imprecisely. Titrations are carried out with insufficient care or data recorded imprecisely. The value of the equilibrium constant is not in the expected range.
Investigate how pH changes when a weak acid reacts with a strong base or when a strong acid reacts with a weak base For example: Determine the pH curve for ethanoic acid reacting with sodium hydroxide	2 marks: All areas of the task are carried out competently. The apparatus is used correctly. The pH values are recorded correctly. The pH changes are in the expected range. 1 mark: One of the areas of the task is performed poorly. The apparatus is used incorrectly OR The pH values are recorded incorrectly OR The pH changes are not in the expected range. 0 marks: At least two of the areas of the task are performed poorly. The apparatus is used incorrectly. The pH values are recorded incorrectly. The pH changes are not in the expected range.

Practical Skills Assessment: A2 Organic Chemistry

Task and possible context	Specific marking guidance
Prepare a solid organic compound For example: The preparation of aspirin	2 marks: All areas of the task are carried out competently. The quantities of reagents are appropriate for the preparation. The apparatus set-up for the preparation is safe and appropriate. The experiment is carried out safely and produces an appropriate quantity and quality of product. 1 mark: One of the areas of the task is performed poorly. The quantities of reagents are inappropriate for the preparation OR The apparatus set-up for each experiment is unsafe or inappropriate OR The experiment is carried out with insufficient care or the yield is poor. 0 marks: At least two of the areas of the task are performed poorly. The quantities of reagents are inappropriate for the preparation. The apparatus set-up for each experiment is unsafe or inappropriate. The experiment is carried out with insufficient care or the yield is poor.
Purify an organic solid For example: The recrystallisation of impure benzenecarboxylic acid from hot water	2 marks: All areas of the task are carried out competently. The quantity of solvent is appropriate. The recrystallisation process is carried out safely and with due care. The quantity and quality of recrystallised product are both appropriate. 1 mark: One of the areas of the task is performed poorly. The quantity of solvent is inappropriate OR The recrystallisation process is carried out with insufficient care OR Either the quantity or quality of recrystallised product is inappropriate. 0 marks: At least two of the areas of the task are performed poorly. The quantity of solvent is inappropriate. The recrystallisation process is carried out with insufficient care. Either the quantity or quality of recrystallised product is inappropriate
Test the purity of an organic solid For example: Determine the melting point of benzenecarboxylic acid	2 marks: All areas of the task are carried out competently. The quantity used and the preparation of the solid are appropriate (e.g. dry, powder). The apparatus set-up is safe and appropriate. Heating is carried out with due care and only as long as necessary, giving an accurate value for the melting point. 1 mark: One of the areas of the task is performed poorly. Either the quantity used or the preparation of the solid is inappropriate OR The apparatus set-up is unsafe or inappropriate OR Heating is longer than necessary and the m.p. is inaccurate. 0 marks: At least two of the areas of the task are performed poorly. Either the quantity used or the preparation of the solid is inappropriate. The apparatus set-up is unsafe or inappropriate. Heating is longer than necessary and the m.p. is inaccurate.